ITI – PLSQL Advanced Lab 1

Mohammad Tarek Omar – Data Management

1. **Create and invoke the ADD_LOC procedure and consider the results.**

   a) **Create a procedure called ADD_LOC to insert a new Location into the LOCATIONS Provide the LOCATION_ID , STREET_ADDRESS, POSTAL_CODE , CITY, STATE_PROVINCE, COUNTRY_ID parameters.**

   b) **Compile the code; invoke (call) the procedure. Query the Locations table to view the results.**

   c) **Handle Error for the Invalid Country IDs**

```oracle_plsql
-- 1.
create or replace procedure ADD_LOC (Loc_id number, Str_addr varchar2, post_code number, city varchar2, state_prov varchar2, country_id varchar2)
is
begin
   insert into Locations (LOCATION_ID, STREET_ADDRESS, POSTAL_CODE, CITY, STATE_PROVINCE, COUNTRY_ID)
   values (Loc_id, Str_addr, post_code, city, state_prov, country_id);
end;
---------------
declare
   wrng_count_id_exeption exception;
   pragma exception_init(wrng_count_id_exeption, -2291);
begin
ADD_LOC(3300, 'street 15', '265478', '15th of may', 'Cairo', 'EG');
exception when wrng_count_id_exeption then
      dbms_output.put_line('PLease enter a valid country_id');
end;
select * from Locations where location_id = 3300
----------------------------------------
```

Script Output

Data Grid | Script Output

Output | Grid 1 | Environment

```
        3300 street 15                          265478
15th of may                    Cairo                EG

1 row selected.
```

```oracle_plsql
set serveroutput on
declare
   wrng_count_id_exeption exception;
   pragma exception_init(wrng_count_id_exeption, -2291);
begin
ADD_LOC(3800, 'street 15', '265478', '15th of may', 'Cairo', 'd5');
exception when wrng_count_id_exeption then
      dbms_output.put_line('PLease enter a valid country_id');
end;
select * from Locations where location_id = 3800
----------------------------------------
```

Script Output

Data Grid | Script Output

Output | Environment

```
PLease enter a valid country_id
PL/SQL procedure successfully completed.
```

2. Create and Invoke the Query_loc Function to display the data for a certain region from Locations, Countries, regions tables in the following format " Region Name , Country Name , LOCATION_ID , STREET_ADDRESS , POSTAL_CODE , CITY " Pass Location ID as an input parameter

Hint : concat them in a single character variable and return it

```
create or replace function Query_loc  (location_idd number)
return varchar2
is
   result varchar2(1000);
begin
   select r.region_name || ', ' ||
          c.country_name || ', ' ||
          l.location_id || ', ' ||
          l.street_address || ', ' ||
          l.postal_code || ', ' ||
          l.city
  into result
   from locations l
   join countries c on l.country_id = c.country_id
   join regions r   on c.region_id = r.region_id
   where  l.location_id = location_idd;
   return result;
end;
show errors
----------------------
set serveroutput on;
declare
result varchar2(1000);
begin
   result := Query_loc(3200);
   dbms_output.put_line(result);
end;
```


```
set serveroutput on;
declare
result varchar2(1000);
begin
   result := Query_loc(3200);
   dbms_output.put_line(result);
end;
```


Script Output

Data Grid | Script Output

Output | Environment

```
Americas, Mexico, 3200, Mariano Escobedo 9991, 11932, Mexico City
PL/SQL procedure successfully completed.
```

3. Create and invoke the GET_LOC function to return the street address, city for a specified LOCATION_ID.

Hint : use out parameters for the city

```
-- 3.
create or replace function GET_LOC (location_idd number, cityy out varchar2)
return varchar2
is
  str_add varchar2(100);
begin
   select street_address, city
   into  str_add, cityy
   from locations
   where location_id = location_idd;
   return  str_add;
end;
---------------------
set serveroutput on;
declare
   city varchar(100);
   street varchar(100);
begin
   street := GET_LOC(3100, city);
   dbms_output.put_line(street);
   dbms_output.put_line(city);
end;
```


```
set serveroutput on;
declare
   city varchar(100);
   street varchar(100);
begin
   street := GET_LOC(3100, city);
   dbms_output.put_line(street);
   dbms_output.put_line(city);
end;
```


Script Output

Data Grid | Script Output

Output | Environment

```
Pieter Breughelstraat 837
Utrecht
PL/SQL procedure successfully completed.
```

4. Create a function called GET_ANNUAL_COMP to return the annual salary computed from an employee's monthly salary and commission passed as parameters. Use the following basic formula to calculate the annual salary:

(Salary*12) + (commission_pct*salary*12)

Use the function in a SELECT statement

Hint: Function call prototype **GET_ANNUAL_COMP(7000, 0.15)**

```
-- 4.
create or replace function GET_ANNUAL_COMP(salaryy number, comm number)
return number
is
   result number(8, 2);
begin
   result := (salaryy*12) + (comm*salaryy*12);
   return result;
end;
-----------
select GET_ANNUAL_COMP(7000, 0.15) from employees
```

Script Output

Data Grid | Script Output

Output | Grid 1 | Environment

```
GET_ANNUAL_COMP(7000,0.15)
--------------------------
                    96600
```

5. a- add RETIRED NUMBER(1) column to employees table using alter

```
84  ----------------------------
85  -- 5.
86  alter table employees add retired number(1);
87
```

b- Create and call

CHECK_RETIRED FUNCTION(V_EMP_ID NUMBER, V_MAX_HIRE_YEAR NUMBER) RETURN Number;

that will return 1 if employee has passed no of years >= V_MAX_HIRE_YEAR, return 0 for otherwise

```
85  -- 5.
86  alter table employees add retired number(1);
87
88  create or replace function CHECK_RETIRED(V_EMP_ID NUMBER, V_MAX_HIRE_YEAR NUMBER)
89  return number
90  is
91      hire_datee date;
92      years number;
93  begin
94      select hire_date into hire_datee from employees
95      where employee_id = V_EMP_ID;
96
97
98      years := trunc(months_between(sysdate, hire_datee) / 12);
99      if years >= V_MAX_HIRE_YEAR then
100         return 1;
101     else
102         return 0;
103     end if;
104 end;
105 show errors
```

Script Output

Data Grid | Script Output

Output | Environment

```
Function created.
No errors.
```

c- create anonymous block to update the emp with set retired = 1 if this employee will retired

```oracle
select * from employees;
declare
   retiredd number;
   cursor emp_ret is select employee_id from employees;
begin
   for roww in  emp_ret loop
      retiredd := CHECK_RETIRED(roww.employee_id, 20);
      if retiredd = 1 then
         update employees set retired = 1
         where employee_id = roww.employee_id;
      else
         update employees set retired = 0
         where employee_id = roww.employee_id;
      end if;
   end loop;
end;
select * from employees;
```

Script Output

Data Grid | Script Output

Output | Grid 1 | Grid 2 | Environment

```
                    205            110            1


107 rows selected.
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	RETIRED
172	Elizabeth	Bates	EBATES	011.44.1343.529268	24-MAR-07	SA_REP	7300	.15	148	80	
173	Sundita	Kumar	SKUMAR	011.44.1343.329268	21-APR-08	SA_REP	6100	.1	148	80	
174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	.3	149	80	
175	Alyssa	Hutton	AHUTTON	011.44.1644.429266	19-MAR-05	SA_REP	8800	.25	149	80	
176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	24-MAR-06	SA_REP	8600	.2	149	80	
177	Jack	Livingston	JLIVINGS	011.44.1644.429264	23-APR-06	SA_REP	8400	.2	149	80	
178	Kimberely	Grant	KGRANT	011.44.1644.429263	24-MAY-07	SA_REP	7000	.15	149		
179	Charles	Johnson	CJOHNSON	011.44.1644.429262	04-JAN-08	SA_REP	6200	.1	149	80	
180	Winston	Taylor	WTAYLOR	650.507.9876	24-JAN-06	SH_CLERK	3200		120	50	
181	Jean	Fleaur	JFLEAUR	650.507.9877	23-FEB-06	SH_CLERK	3100		120	50	

Output | Grid 1 | Grid 2 | Environment

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	RETIRED
110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200		108	100	1
111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700		108	100	1
112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800		108	100	0
113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900		108	100	0
114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000		100	30	1
115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100		114	30	1
116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900		114	30	0
117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800		114	30	1
118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600		114	30	0
119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500		114	30	0